

Exam Date & Time: 27-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Ethical Hacking and Cyber Security [CSE-4066]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Identify and explain different types of active attacks and passive attacks. (5)
 - A)
 - B) Consider you are working in a company named XYZ solutions as ethical hacker wherein you need to perform some penetration testing. Demonstrate the tool which can be used for this task along with the sample code. (5)
- 2) Consider you are being given the task of gathering information during the enumeration process. Examine the different types of enumeration in detail. (5)
 - A)
 - B) Examine the Linux vulnerability with an example tool demonstration. (5)
- 3) Illustrate how web applications interact with databases, including the use of SQL queries and data retrieval. Explain how this interaction can be exploited if not secured properly. (5)
 - A)
 - B) Discuss the following:
 - i) ODBC
 - ii) ActiveX Data Objects (ADO)(5)
- 4) Demonstrate how phishing attacks target specific individuals or organizations and discuss the impact these attacks can have on personal data and organizational security. (5)
 - A)
 - B) Explain the concept of a Trojan horse in cybersecurity and discuss how attackers use backdoors to gain unauthorized access to a system. (5)
- 5) Examine the importance of two-factor authentication in reducing real time credit card fraud in mobile transactions. What are some challenges in implementing this security feature? (5)
 - A)
 - B) Examine with a real time example how cryptographic techniques, such as hashing and encryption, help secure sensitive data in web applications. (5)

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